

Name: Jessica John Osta

Register no.: 2460376

Class: 5BTCS B

AI Application: Ethical AI Document Summarizer and Knowledge Assistant (Chatbot)

Problem Statement:

Professionals and students struggle to quickly extract accurate, actionable insights from massive volumes of complex textual data. Current AI tools often risk leaking sensitive information, misinterpreting context, or introducing factual inaccuracies (hallucinations) without user oversight.

Data Required:

- **Input Data:** Target documents (PDFs, Word docs, text files, or URLs) uploaded by the user.
- **Contextual Data:** User-entered text prompts, questions, or specific summarization instructions.
- **System Data:** User interaction history within the specific chat session.

Expected Output:

- High-fidelity, concise document summaries (executive summaries, bulleted key takeaways, or action items).
- Context-aware, natural language responses to user questions, strictly grounded in the uploaded document's text.

Ethical Principles:

- **Privacy:** Absolute data protection ensuring user-uploaded documents are encrypted in transit and at rest, and are never used to train future public foundation models.
- **Transparency:** The system clearly cites or highlights the exact sections of the original document used to generate a specific summary or answer.
- **Fairness:** The underlying Language Model is regularly audited to ensure it summarizes content objectively, without amplifying systemic biases or marginalizing specific perspectives present in the text.
- **Safety & Reliability:** Implementation of rigorous guardrails to eliminate "hallucinations" (fabricated facts) and prevent the chatbot from generating toxic, harmful, or misleading content.

- **Accountability:** Establishing clear ownership of the system's performance, maintaining system logs for ethical audits, and providing a clear mechanism for users to flag incorrect outputs.

Ethical Risk:

Data Privacy Violation and Intellectual Property Leakage. When users upload confidential documents (such as proprietary corporate strategies, legal contracts, or unpublished research papers), there is a significant risk that the data could be cached, stored insecurely, or fed into public training datasets. This could result in sensitive information being exposed to unauthorized third parties or leaked through another user's prompt down the line.

Mitigation Strategy:

Implement an **Enterprise-Grade Privacy Boundary and Data Zero-Retention Policy.**

1. **Local Embeddings & Isolated Architecture:** Utilize a secure Vector Database that isolates each user's data into a sandboxed environment. Documents are converted into mathematical vectors (embeddings) locally or within an encrypted cloud container.
2. **Zero-Data Retention (ZDR) API Agreements:** Connect only to AI model providers that explicitly guarantee through APIs that user inputs are processed solely in real-time memory and are **never** stored, reviewed by humans, or used for model training.
3. **Session-End Purging:** Automatically purge all uploaded text and vectors the moment the user closes the session, leaving no residual data trail